

Phase Transitions in Sparse Feature Representations

Reuven Balkin

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Abstract

These notes derive the analytical results underlying the toy superposition model notebooks. We follow the setup of Elhage et al. (2022) and study a minimal one-hidden-unit autoencoder ($n = 2$ features, $m = 1$ latent dimension) trained on sparse, independently distributed inputs. Section 1 treats the linear model and shows that superposition cannot occur: the only minima of the expected loss are the single-feature solutions $\theta = 0$ and $\theta = \pi/2$, and the non-trivial critical point is always a saddle. Section 2 introduces a ReLU nonlinearity and derives a three-phase structure — including a genuine superposition minimum at $\theta = 3\pi/4$ — via exact closed-form expressions for the expected loss.

1 Linear model

Our setup follows a toy model presented in 'Toy Models of Superposition', where we attempt to reconstruct a vector $x \in \mathbb{R}^n$ by first projecting him to a lower-dimensional space $\mathbb{R}^m$. We define

$$h = wx, \quad (1)$$

$$x = w^T h + b, \quad (2)$$

$$x' = w^T wx + b. \quad (3)$$

The problem does not have a preferred basis (for any solution w , Ow is also a solution where O is an orthogonal rotation, therefore the sensible object to study is $w^T w$). We start with the simplest problem $n = 2, m = 1$. We parameterize

$$w = (c_\theta, s_\theta), \quad (4)$$

$$w^T w = \begin{pmatrix} c_\theta^2 & \frac{1}{2}s_{2\theta} \\ \frac{1}{2}s_{2\theta} & s_\theta^2 \end{pmatrix}. \quad (5)$$

The network is trained on the set x_{Ji} where $J = 1, \dots, k$ and $i = 1, 2$. Each x_{Ji} is an i.i.d with probability $1 - p_i$ to be 0, and probability p_i to be distributed uniformly between 0 and 1. The loss function is a weighted MSE, which can be written as

$$\mathcal{L} = \frac{1}{2k} \sum_J \sum_i I_i (x_{Ji} - x'(x_{Ji}))^2 \equiv \frac{1}{2m} \sum_J \mathcal{L}_J \quad (6)$$

where I_i are the weights for each of the features. For $k \gg 1$, the loss function is a narrowly-distributed variable, and we can consider its mean in order to study the phases of the system. The fundamental i.i.d variables satisfy

$$E[x_{Ji}] = \frac{p_i}{2}, \quad E[x_{Ji}^2] = \frac{p_i}{3}. \quad (7)$$

We find

$$\begin{aligned} E[\mathcal{L}] &= \frac{1}{6} I_1 s_\theta^4 p_1 + \frac{1}{6} I_2 c_\theta^4 p_2 + \frac{s_{2\theta}^2}{24} \sum_i I_i p_i + \frac{1}{2} \sum_i I_i b_i^2 \\ &\quad - \frac{s_{2\theta}}{8} (I_1 s_\theta^2 + I_2 c_\theta^2) p_1 p_2 - \frac{1}{2} (I_1 s_\theta^2 b_1 p_1 + I_2 c_\theta^2 b_2 p_2) + \frac{s_{2\theta}}{4} \sum_i I_i p_i b_i \end{aligned} \quad (8)$$

We find the solutions for b in the minimum to be,

$$b_1^* = \frac{s_\theta}{2}(p_1 s_\theta - p_2 c_\theta) \quad (9)$$

$$b_2^* = \frac{c_\theta}{2}(p_2 c_\theta - p_1 s_\theta), \quad (10)$$

note that $\frac{\partial^2 E[\mathcal{L}]}{\partial b_i^2} = I_i > 0$. Therefore the loss function become a function of θ alone, namely

$$E[\mathcal{L}]|_{b^*} = \frac{1}{6}(I_1 p_1 s_\theta^4 + I_2 p_2 c_\theta^4) + \frac{s_{2\theta}^2}{24}(I_1 p_2 + I_2 p_1) - \frac{(I_1 s_\theta^2 + I_2 c_\theta^2)(p_1^2 s_\theta^2 + p_2^2 c_\theta^2)}{8}. \quad (11)$$

and the first and second derivatives

$$\left. \frac{\partial E[\mathcal{L}]}{\partial \theta} \right|_{b^*} = \frac{s_{2\theta}}{6}(I_1 p_1 - I_2 p_2) - \frac{s_{2\theta}}{4}(I_1 p_1^2 s_\theta^2 - I_2 p_2^2 c_\theta^2) \quad (12)$$

$$\left. \frac{\partial^2 E[\mathcal{L}]}{\partial \theta^2} \right|_{b^*} = \frac{2c_{2\theta}}{6}(I_1 p_1 - I_2 p_2) - \frac{2c_{2\theta}}{4}(I_1 p_1^2 s_\theta^2 - I_2 p_2^2 c_\theta^2) - \frac{s_{2\theta}^2}{4}(I_1 p_1^2 + I_2 p_2^2) \quad (13)$$

Let us analyze the possible stationery points. First we have trivial solution at $\theta = 0, \pi/2$, for which

$$\left. \frac{\partial^2 E[\mathcal{L}]}{\partial \theta^2} \right|_{b^*, \theta=0} = \frac{1}{3}(I_1 p_1 - I_2 p_2) + \frac{1}{2}I_2 p_2^2 \quad (14)$$

$$\left. \frac{\partial^2 E[\mathcal{L}]}{\partial \theta^2} \right|_{b^*, \theta=\pi/2} = -\frac{1}{3}(I_1 p_1 - I_2 p_2) + \frac{1}{2}I_1 p_1^2 \quad (15)$$

Thus, we have two clear cases where if

$$I_1 p_1 - I_2 p_2 < -(3/2)I_2 p_2^2 \rightarrow \theta = \pi/2, \quad (16)$$

$$I_1 p_1 - I_2 p_2 > (3/2)I_1 p_1^2 \rightarrow \theta = 0, \quad (17)$$

For $-(3/2)I_2 p_2^2 < I_1 p_1 - I_2 p_2 < (3/2)I_1 p_1^2$ both are minima, with one being the deeper the two. We note however that this region becomes tighter with scarcity, since the interval edges scale like $\mathcal{O}(p^2)$. Indeed, the problem becomes much simpler if we neglect $\mathcal{O}(p^2)$ terms, in which case $\max[I_1 p_1, I_2 p_2]$ selects which feature is represented.

There is an additional non-trivial solution when $s_{2\theta} \neq 0$,

$$c_{\theta_*}^2 = \frac{I_1 p_1^2 - \frac{2}{3}(I_1 p_1 - I_2 p_2)}{I_2 p_2^2 + I_1 p_1^2} \quad (18)$$

The second derivative at θ_* is given by

$$\left. \frac{\partial^2 E[\mathcal{L}]}{\partial \theta^2} \right|_{b^*, \theta=\theta_*} = -\frac{s_{2\theta}^2}{4}(I_1 p_1^2 + I_2 p_2^2) < 0. \quad (19)$$

Therefore the only minima are 0 and $\pi/2$, and superposition does not occur.

Phase boundary

To determine which of the two minima is the global one, we evaluate $E[\mathcal{L}]|_{b^*}$ at each. At $\theta = 0$ the network represents only feature 1 ($w^T w = \text{diag}(1, 0)$), so the loss reduces entirely to the residual on feature 2:

$$E[\mathcal{L}]|_{b^*, \theta=0} = I_2 p_2 \left(\frac{1}{6} - \frac{p_2}{8} \right). \quad (20)$$

Similarly, at $\theta = \pi/2$ only feature 2 is represented and

$$E[\mathcal{L}]|_{b^*, \theta=\pi/2} = I_1 p_1 \left(\frac{1}{6} - \frac{p_1}{8} \right). \quad (21)$$

Feature 1 is represented ($\theta = 0$ is the global minimum) when the first expression is smaller, giving the exact phase boundary

$$\frac{I_2}{I_1} = \frac{p_1(1 - \frac{3p_1}{4})}{p_2(1 - \frac{3p_2}{4})}. \quad (22)$$

2 Non-Linear model

$$h = wx, \quad x' = \text{ReLU}[w^T h + b] = \text{ReLU}[w^T w x + b]. \quad (23)$$

We again take $n = 2$, $m = 1$, with the same parameterization

$$w = (c_\theta, s_\theta), \quad w^T w = \begin{pmatrix} c_\theta^2 & \frac{1}{2} s_{2\theta} \\ \frac{1}{2} s_{2\theta} & s_\theta^2 \end{pmatrix}. \quad (24)$$

The network is trained on $\{x_{Ji}\}$ with $J = 1, \dots, k$ and $i = 1, 2$. Each x_{Ji} is i.i.d.: zero with probability $1 - p_i$, Uniform $[0, 1]$ with probability p_i . The loss function is a weighted MSE,

$$\mathcal{L} = \frac{1}{2k} \sum_J \sum_i I_i (x_{Ji} - x'_{Ji})^2, \quad (25)$$

with moments $E[x_{Ji}] = p_i/2$ and $E[x_{Ji}^2] = p_i/3$.

For $k \gg 1$, $\mathcal{L}$ concentrates around its mean. We decompose $E[\mathcal{L}_J]$ according to which features are active, absorbing the probability weights into each term:

$$E[\mathcal{L}_J] = A + B + C + D. \quad (26)$$

2.1 Both features inactive

($x_1 = x_2 = 0$, probability $(1 - p_1)(1 - p_2)$). When both inputs vanish, $x'_i = \text{ReLU}(b_i)$, so

$$A = (1 - p_1)(1 - p_2) \sum_i I_i \text{ReLU}(b_i)^2. \quad (27)$$

This piece is leading in the sparse-features regime and is independent of θ ; if both features are zero, the best reconstruction would be for $b_i \leq 0$. Thus, this term in the loss function pushes b_i the non-positive values.

2.2 Only feature 1 active

In this case, $x_1 \sim \text{Unif}[0, 1]$, $x_2 = 0$, with probability $(1 - p_2)p_1$. The reconstructions reduce to $x'_1 = \text{ReLU}(c_\theta^2 x_1 + b_1)$ and $x'_2 = \text{ReLU}(\frac{s_{2\theta}}{2} x_1 + b_2)$. We introduce the single-feature auxiliary integrals

$$F_n(\alpha, \beta) \equiv \int_0^1 \text{ReLU}(\alpha x + \beta)^n dx = \frac{\text{ReLU}(\alpha + \beta)^{n+1} - \text{ReLU}(\beta)^{n+1}}{\alpha(n+1)}, \quad (28)$$

$$G(\alpha, \beta) \equiv \int_0^1 x \text{ReLU}(\alpha x + \beta) dx = \frac{1}{\alpha} \left[\frac{\text{ReLU}(\alpha + \beta)^3 - \text{ReLU}(\beta)^3}{3\alpha} - \beta F_1(\alpha, \beta) \right], \quad (29)$$

and find

$$B = (1 - p_2)p_1 \left[I_1 \left(\frac{1}{3} - 2G(c_\theta^2, b_1) + F_2(c_\theta^2, b_1) \right) + I_2 F_2\left(\frac{s_{2\theta}}{2}, b_2\right) \right]. \quad (30)$$

2.3 Only feature 2 active

By symmetry ($1 \leftrightarrow 2$, $c_\theta \leftrightarrow s_\theta$):

$$C = (1 - p_1)p_2 \left[I_1 F_2\left(\frac{s_{2\theta}}{2}, b_1\right) + I_2 \left(\frac{1}{3} - 2G(s_\theta^2, b_2) + F_2(s_\theta^2, b_2) \right) \right]. \quad (31)$$

2.4 No superposition

Consider the $\theta = 0$ phase. The loss function is minimized with the choice $b_1 = 0$, and one ends up with

$$(A + B + C) \Big|_{\theta=0, b_1=0} = I_2 \left[\text{ReLU}^2(b_2) + p_2 \left(\frac{1}{3} - \text{ReLU}(b_2) \right) \right] + \mathcal{O}(p^2) \quad (32)$$

The loss is then minimized for $b_2 = p_2/2$, and we find

$$(A + B + C) \Big|_{\theta=0, b_1=0, b_2=p_2/2} = \frac{I_2 p_2}{3}. \quad (33)$$

By symmetry

$$(A + B + C) \Big|_{\theta=\pi/2, b_2=0, b_1=p_1/2} = \frac{I_1 p_1}{3}. \quad (34)$$

2.5 Superposition

Let us consider $\theta = 3\pi/4$. In this case, the calculation is more involved since different values of b_1, b_2 lead to different ReLU's being activated. Nevertheless, after a straight forward calculation, one finds that the loss function is minimized for $b_1, b_2 \rightarrow 0^+$, and

$$(A + B + C) \Big|_{\theta=3\pi/4, b_1=b_2=0} = \frac{I_1 p_1 + I_2 p_2}{12} - \mathcal{O}(p^2). \quad (35)$$

which allows us to determine the phase structure

$$\theta = \begin{cases} 0 & I_2 p_2 < (I_1 p_1)/3 \\ 3\pi/4 & (I_1 p_1)/3 < I_2 p_2 < 3I_1 p_1 \\ \pi/2 & 3I_1 p_1 < I_2 p_2 \end{cases}. \quad (36)$$

2.6 Both features active

($x_1, x_2 \sim \text{Unif}[0, 1]$, probability $p_1 p_2$). Now both inputs enter the reconstruction, coupling x_1 and x_2 . Integrating over both requires the double auxiliary integrals

$$\begin{aligned} \mathcal{F}_n(\alpha, \gamma, \beta) &\equiv \int_0^1 F_n(\alpha, \gamma y + \beta) dy \\ &= \frac{\text{ReLU}(\alpha + \gamma + \beta)^{n+2} - \text{ReLU}(\gamma + \beta)^{n+2} - \text{ReLU}(\alpha + \beta)^{n+2} + \text{ReLU}(\beta)^{n+2}}{\alpha \gamma (n+1)(n+2)}, \end{aligned} \quad (37)$$

$$\begin{aligned} \mathcal{H}(\alpha, \gamma, \beta) &\equiv \int_0^1 y F_1(\alpha, \gamma y + \beta) dy \\ &= \frac{\text{ReLU}(\alpha + \gamma + \beta)^3 - \text{ReLU}(\gamma + \beta)^3}{6\alpha\gamma} - \frac{\text{ReLU}(\alpha + \gamma + \beta)^4 - \text{ReLU}(\alpha + \beta)^4 - \text{ReLU}(\gamma + \beta)^4 + \text{ReLU}(\beta)^4}{24\alpha\gamma^2}, \end{aligned} \quad (38)$$

$$\mathcal{G}(\alpha, \gamma, \beta) \equiv \int_0^1 G(\alpha, \gamma y + \beta) dy = \frac{1}{\alpha} \left[\mathcal{F}_2(\alpha, \gamma, \beta) - \beta \mathcal{F}_1(\alpha, \gamma, \beta) - \gamma \mathcal{H}(\alpha, \gamma, \beta) \right], \quad (39)$$

giving

$$\begin{aligned} D &= p_1 p_2 \left[I_1 \left(\frac{1}{3} - 2\mathcal{G}(c_\theta^2, \frac{s_{2\theta}}{2}, b_1) + \mathcal{F}_2(c_\theta^2, \frac{s_{2\theta}}{2}, b_1) \right) \right. \\ &\quad \left. + I_2 \left(\frac{1}{3} - 2\mathcal{G}(s_\theta^2, \frac{s_{2\theta}}{2}, b_2) + \mathcal{F}_2(s_\theta^2, \frac{s_{2\theta}}{2}, b_2) \right) \right]. \end{aligned} \quad (40)$$

Combining all four contributions, the expected loss is

$$\boxed{E[\mathcal{L}] = \frac{1}{2}(A + B + C + D)}. \quad (41)$$

Summary

The two models studied here bracket the role of the nonlinearity. In the linear case the expected loss is a polynomial in θ whose only minima are at $\theta = 0$ and $\theta = \pi/2$; the non-trivial stationary point is always a saddle, so superposition is impossible. The ReLU breaks this symmetry: the reconstruction of a positively-activated feature and a negatively-activated one are decoupled by the nonlinearity, opening up the antipodal configuration $\theta = 3\pi/4$ as a genuine minimum. The resulting phase boundary depends on I_i and p_i only through their product $q_i = I_i p_i$ at leading order in sparsity, collapsing the four-parameter family to a two-dimensional phase diagram. Corrections at finite p_i are encoded in the D term and are explored numerically in the companion notebooks.

References

- [1] Elhage et al., *Toy Models of Superposition*, Transformer Circuits Thread (2022).
https://transformer-circuits.pub/2022/toy_model/index.html